Title: Does removing race from eGFR change who an AI misses? Fairness of machine-learning screening for undiagnosed chronic kidney disease in NHANES 2017–March 2020

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Background & Objective: Approximtely 9 in 10 adults with chronic kidney disease (CKD) in the U.S. are unaware of their condition. We aimed to develop and internally validate a two-stage machine learning framework for opportunistic digital screening in primary care, and to audit whether replacing the race-adjusted CKD-EPI 2009 equation with the race-free CKD-EPI 2021 equation changes who is classified as having CKD and whom the model misses.

Methods: Using the pre-pandemic NHANES 2017–March 2020 cohort (N = 7,426 examined adults aged ≥20 years without a prior kidney disease diagnosis and with complete outcome data), we defined undiagnosed CKD as eGFR < 60 mL/min/1.73 m² (computed separately with CKD-EPI 2009 and 2021) or UACR ≥ 30 mg/g in participants never told they had weak or failing kidneys (KIQ022), excluding those with missing serum creatinine, UACR or KIQ022. The primary model, Stage 1 (EHR Zero-Lab), used non-invasive sociodemographic and vital variables for initial triage. Stage 2 (Lab-Enhanced), a secondary analysis, added routine blood biomarkers (glucose, uric acid and total cholesterol), excluding creatinine and UACR, which define the outcome, and blood urea nitrogen, a filtration marker, to prevent direct or indirect target leakage. Race was excluded as a predictor and used only for the post-hoc fairness audit. We trained elastic-net logistic regression (baseline, using the Stage 1 predictors) and XGBoost models on a 70/30 stratified split, weighted by NHANES examination weights (WTMECPRP), with median/mode imputation fitted within the training data; all performance estimates accounted for the complex survey design (SDMVPSU, SDMVSTRA). Models were trained on the CKD-EPI 2021 outcome and evaluated against both definitions. The classification threshold was prespecified to achieve ≥90% sensitivity in out-of-fold training predictions. Fairness was audited by comparing the false negative rate (FNR) and specificity in low-income (PIR < 1.35) non-Hispanic Black participants versus all other participants, under each equation, using Korn–Graubard confidence intervals (Clopper–Pearson when no events occurred) and the MOVER method for between-group differences; AUC confidence intervals used survey bootstrap resampling (1,000 replicates). We also quantified reclassification of CKD status between equations.

Results: The weighted prevalence of undiagnosed CKD was 12.5% (95% CI: 11.5–13.6) with CKD-EPI 2021 and 13.5% (12.3–14.9) with CKD-EPI 2009. Among 7,426 participants, 130 were reclassified between equations (unweighted: 1.75%; weighted: 1.38%), including 14 of 599 low-income non-Hispanic Black participants (PIR <1.35; unweighted: 2.34%; weighted: 1.80%), all of whom became cases under CKD-EPI 2021. In the test set (n=2,229), Stage 1 (Zero-Lab) XGBoost achieved a weighted AUC of 0.736 (95% CI: 0.690–0.783), Stage 2 (Lab-Enhanced, excluding BUN) 0.757 (0.714–0.801), and logistic regression 0.769 (0.725–0.814). Stage 1 XGBoost performed worse than logistic regression (ΔAUC, XGBoost minus logistic regression: −0.033; 95%

CI: −0.061 to −0.005). At the prespecified threshold of 0.0221, test sensitivity was 91.2% and specificity 22.5%; 79.2% of test participants were flagged for confirmatory testing (weighted). The false-negative rate (FNR) was 0% (0/31 cases; 95% CI: 0–11.2) in low-income non-Hispanic Black participants versus 9.4% (5.2–15.3) in other participants under CKD-EPI 2021, and 0% (0/28; 0–12.3) versus 8.3% (4.7–13.3) under CKD-EPI 2009. Specificity under CKD-EPI 2021 was 14.9% (5.6–29.7) versus 22.9% (19.6–26.5). Subgroup-minus-rest differences under CKD-EPI 2021 were −9.4 percentage points for FNR (95% MOVER CI: −15.3 to 2.6) and −8.1 for specificity (−18.0 to 7.2); both intervals included zero.

Limitations & Conclusions: Main limitations include cross-sectional single-measurement data without 90-day KDIGO confirmation, reliance on estimated rather than measured GFR, the small number of cases in the low-income Black subgroup (n = 31 in the test set), and lack of external validation. Replacing CKD-EPI 2009 with CKD-EPI 2021 reclassified few participants and did not change whom the model missed among low-income Black participants; in this subgroup the model missed no cases, but flagged 87% of participants. At a 90%-sensitivity threshold, the race-free zero-lab model flagged most participants and performed worse than logistic regression, so its value for triage is limited; larger subgroup samples and external validation are needed before any clinical use.